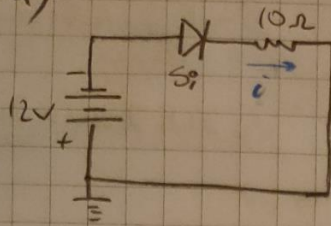
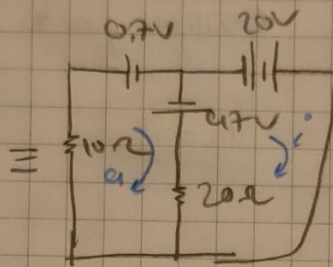
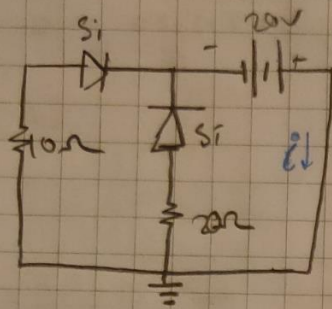


5a)



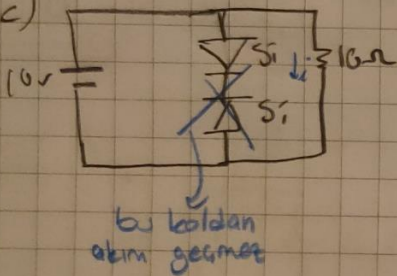
diyot okın ters
deneden okun geçmez
 $i = 0$

5b)



$$\begin{aligned} -20 + 0.7 + 20i &= 0 \\ 0.7 - 0.7 + 30i - 20i &= 0 \\ i &= 0.965A \\ Q &= 0.643A \end{aligned}$$

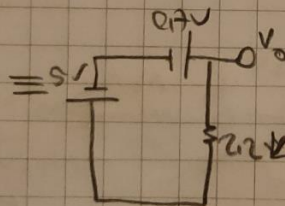
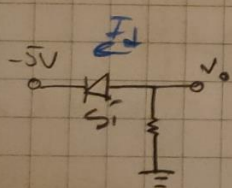
5c)



$$i = (10V / 10\Omega) = \underline{1A}$$

bu koldan
akım geçmez

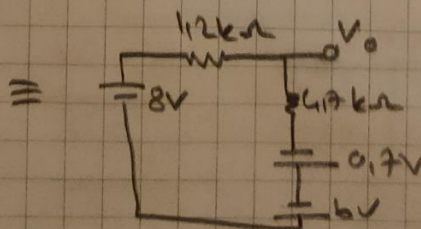
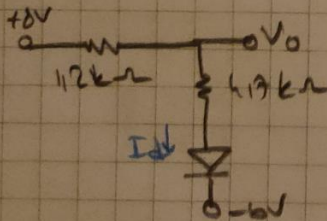
6a)



$$I_d = \frac{5 - 0.7}{20k} = \underline{1.85mA}$$

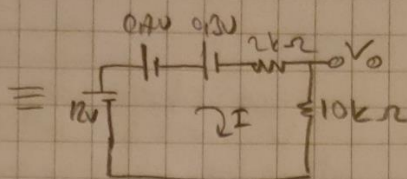
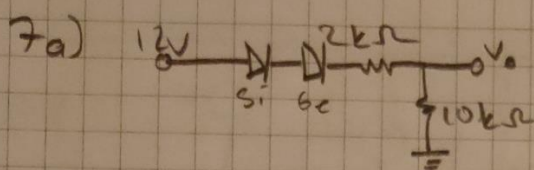
$$V_o = 20k \cdot 1.85mA = \underline{-4.3V}$$

6b)



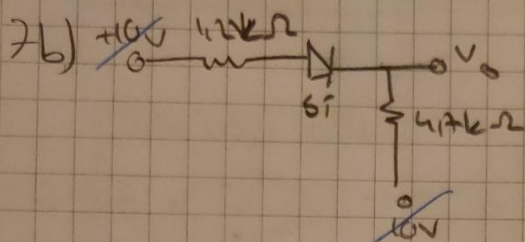
$$I_d = \frac{14 - 0.7}{5.8k} = \underline{2.25mA}$$

$$V_o = 4.7k \cdot 2.25 + 0.7 - 8 = \underline{5.275V}$$



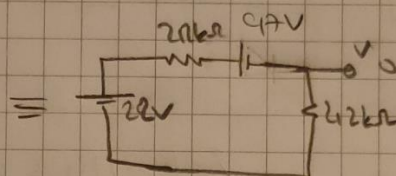
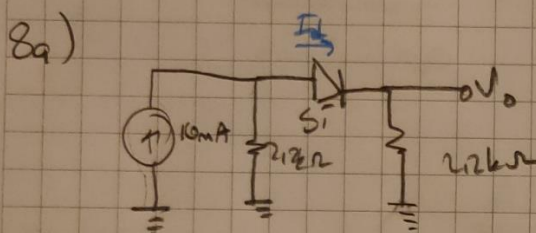
$$I = \frac{11}{12k} = 0,916 \text{ mA}$$

$$V_o = 10 \cdot 0,916 = \underline{\underline{9,16 \text{ V}}}$$



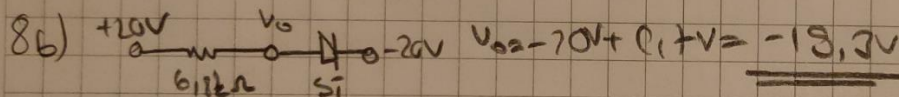
Gerilim farkı yok $\rightarrow$ Akım yok

$$\underline{\underline{V_o = 0}}$$

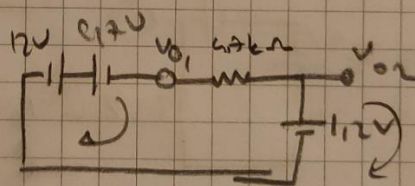
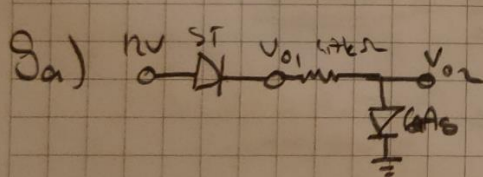


$$I_d = \frac{22 - 0,7}{4k} = 4,86 \text{ mA}$$

$$V_o = 4,86 \cdot 2k = \underline{\underline{9,72 \text{ V}}}$$



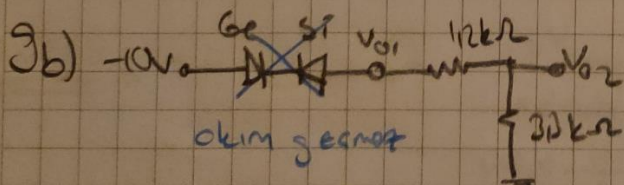
$$V_o = -12V + 0,7V = \underline{\underline{-11,3V}}$$



$$-12 + 0,7 + V_o = 0$$

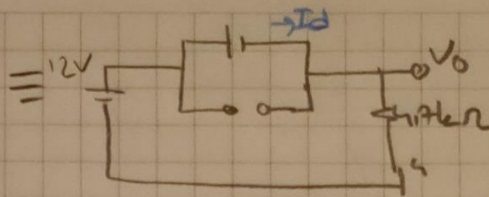
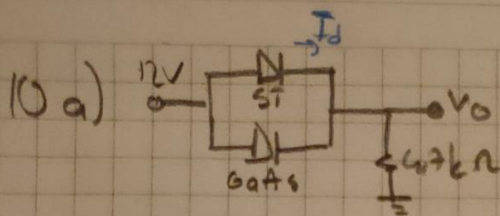
$$\underline{\underline{V_o = 11,3V}}$$

$$\underline{\underline{V_o = 1,2V}}$$



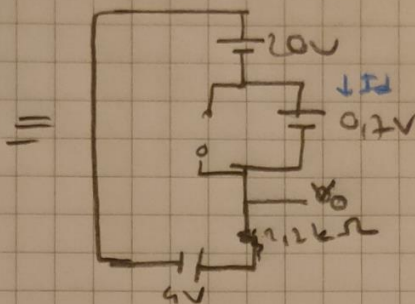
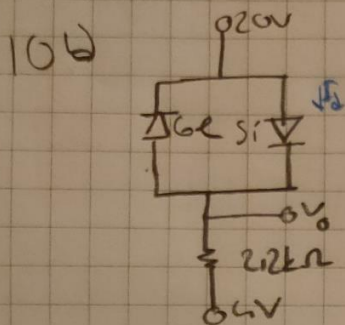
$$\underline{\underline{V_o1 = 0V}}$$

$$\underline{\underline{V_o2 = 0V}}$$



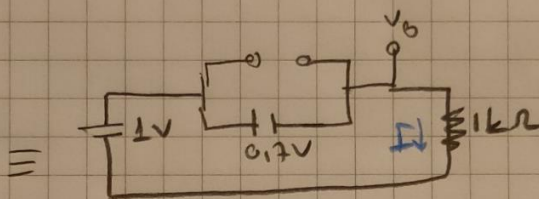
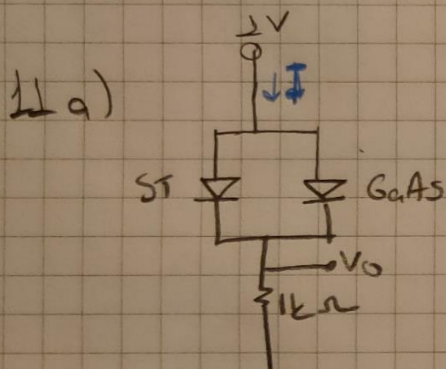
$$I_d = \frac{12 - 0.7}{4.7k} = \underline{\underline{2.4 \text{ mA}}}$$

$$V_o = 12 - 0.7 = \underline{\underline{11.3 \text{ V}}}$$



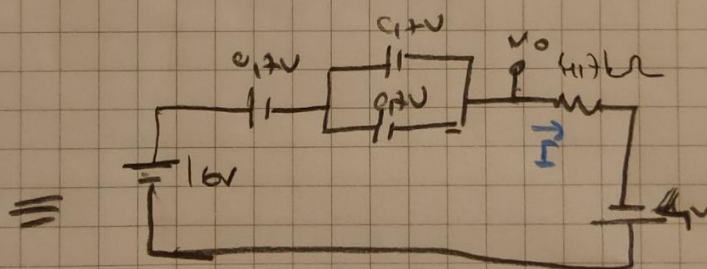
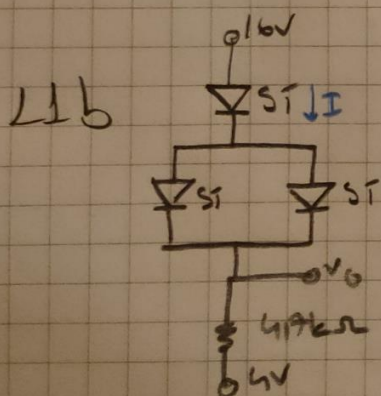
$$V_o = 20 - 0.7 = \underline{\underline{19.3 \text{ V}}}$$

$$I = \frac{19.3 - 4}{2k} = \underline{\underline{6.95 \text{ mA}}}$$



$$V_o = 1 - 0.7 = \underline{\underline{0.3 \text{ V}}}$$

$$I = \frac{0.3}{1k} = \underline{\underline{0.3 \text{ mA}}}$$



$$V_o = 16 - 0.7 - 0.7 - 0.7 = \underline{\underline{14.6 \text{ V}}}$$

$$I = \frac{16 - 0.7 - 0.7 - 0.7 + 4}{4.7k} = \underline{\underline{3.957 \text{ mA}}}$$